

PHASE 15 — Risk-based verification queue

Ye mera recommended big innovation hai.

Instead of:

Every document manually verify.

System:

10,000 records

↓

AI validation

↓

8,400 low risk

1,100 medium risk

500 high risk

↓

Human team focuses on 500

Dashboard:

Verification Queue

● Critical 43

● High 112

● Medium 318

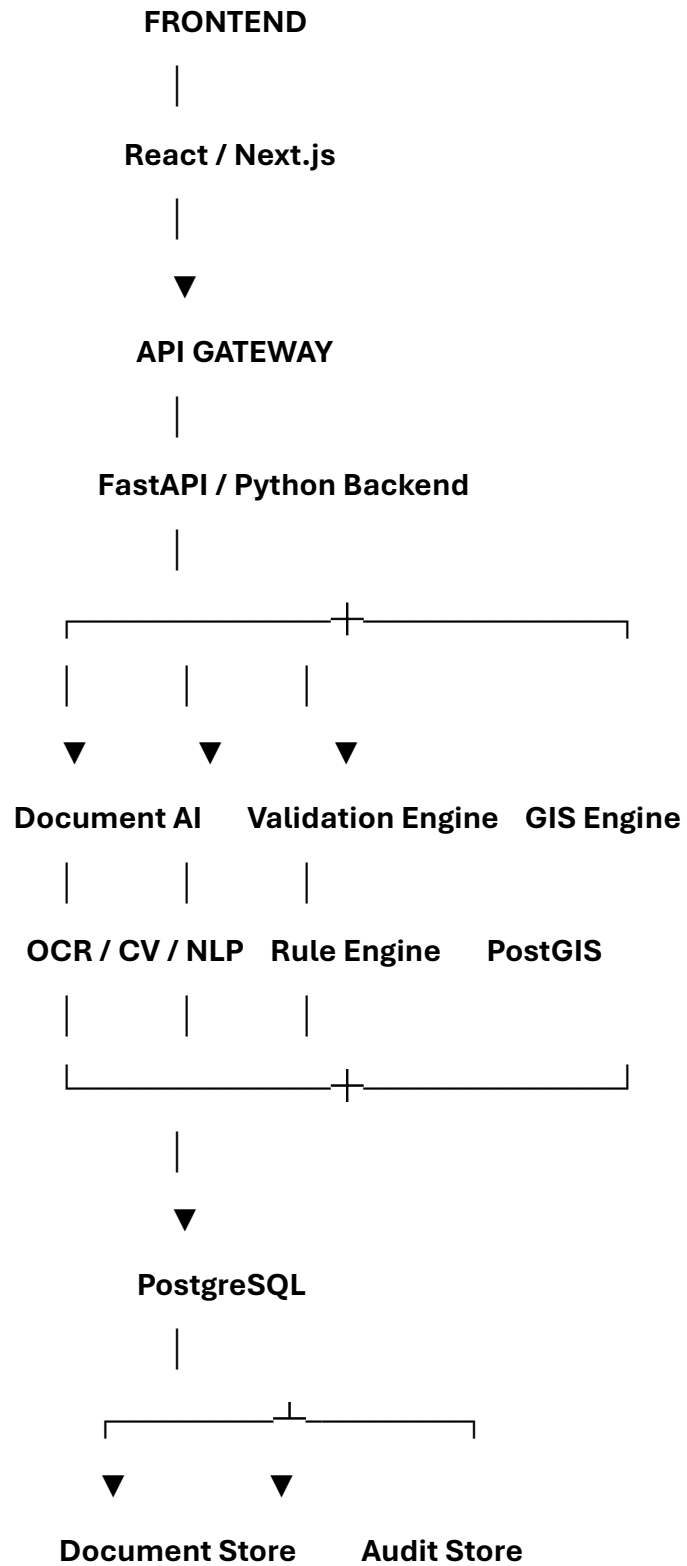
● Low 627

Result:

Manual intervention minimize hota hai, exactly PS ke objective ke according.

✖ Overall Prototype Architecture

Coding stage mein roughly:



Recommended technology stack

PS ke suggested technology ko respect karte hue.

Frontend

React / Next.js

Backend

FastAPI + Python

OCR

Prototype ke liye:

- PaddleOCR / Tesseract baseline
- Indic OCR/HTR model experimentation
- document preprocessing with OpenCV

NLP

- Hugging Face
- Indic NLP
- lightweight LLM/NER layer

CV

- OpenCV
- YOLO/Detectron2 only where actually needed

Database

PostgreSQL + PostGIS

GIS

Leaflet / OpenLayers

Visualization

Plotly

API

REST

Authentication

Prototype:

JWT + RBAC

Storage

Local object storage initially.

Production architecture:

NIC Cloud/MeghRaj or government-approved environment, which the PS itself lists as a suggested option.

 **Prototype mein actual data kaise rakhen?**

Very important.

Government database access ka wait mat karo.

Hum synthetic + publicly available sample-style land records ka controlled demo dataset banayenge.

Example:

Master Record

Record ID: LR001

Owner: Ram Singh

Khasra: 127/2

Khata: 45

Village: Rampur

Area: 2.48 acre

Legacy document

Owner: Ram Singh

Khasra: 127/2

Khata: 45

Area: 2.50 acre

Validation

Owner ✓

Khasra ✓

Khata ✓

Village ✓

Area ⚠

Then second case:

Document:

Owner = Ramesh Singh

Master:

Owner = Ram Singh

GIS:

Area = 2.48

Document:

Area = 5.80

Result:

 **HIGH RISK**

2 conflicts

Ye 2–3 carefully designed cases demo ke liye enough hain.

 **Internal prototype ke liye kitna banana hai?**

MUST HAVE — 100%

Ye complete hona chahiye:

1.

Upload document

2.

Document preview

3.

Image preprocessing

4.

OCR/handwriting extraction

5.

Land-field extraction

6.

Structured record

7.

Field-level confidence

8.

Multi-source validation

9.

Conflict detection

10.

Risk score

11.

Human verification

12.

Evidence viewer

13.

Audit trail

14.

Dashboard

★ SHOULD HAVE

Agar time bache:

15.

GIS map

16.

Duplicate detection

17.

Mutation timeline

18.

Hindi + English

19.

Role-based login

20.

Feedback loop